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FLOW METER

Abstract

A flow meter (**100**) for measuring the flow rate of a fluid characterized by comprising a casing (**1**) having a first end (**101**) into which the fluid enters and a second end (**102**) from which the fluid exits, and within which a first duct (**11**) for receiving the fluid is present when it is at low flow rates; a measuring element (**2**) arranged internally to the first duct (**11**) for measuring the flow within the first duct (**11**);

said flow meter (**100**) being characterized by also comprising: a second duct (**12**), placed inside the casing (**1**), for receiving the fluid at high flow rates a closing element (**3**) which, when the flow rate exceeds an opening threshold, allows the fluid to flow into the second duct (**12**) an elastic element (**4**), connected to the closing element (**3**), for moving the closing element (**3**) from a closing configuration, in which the flow rate does not exceed the opening threshold, not overcoming the resistance of the elastic element (**4**) and the closing element (**3**) closes the second duct (**12**), to an opening configuration in which the flow rate exceeds the opening threshold, overcoming the resistance of the elastic element (**4**), and the closing element (**3**) opens the second duct (**12**).

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Background/Summary

RELATED APPLICATION

[0001] This application claims the benefit of priority of Italy Patent Application No. 102024000004981 filed on Mar. 6, 2024, the contents of which are incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

[0002] The present invention concerns a flow meter with bypass.

[0003] The invention fits into the technical field of flow measurement instruments, in particular in the technical field of flow meters for measuring the flow rate of a liquid.

[0004] There is a wide range of flow measurement instruments on the market and in particular flow meters for measuring the flow or mass flow rate of a liquid or a gas.

[0005] Flow meters have considerable relevance in the market since, in any sector of modern industry, flow measurement has become essential in process controls.

[0006] Advances in technology currently provide a notable variety of spring, ultrasonic, turbine, rotary blade, volumetric, vortex, magnetic, Pitot tube, calibrated diaphragm, etc. flow meters.

[0007] However, although flow meters have become increasingly precise and robust over time, they still have some limitations.

[0008] In particular, systems capable of measuring the flow rate of a liquid at both low and high flow rates are extremely complex and expensive.

[0009] In more detail, the cheapest and most widely used flow meters measure the flow rate either at low flow rates, or at high flow rates but not in both situations.

[0010] For example, there are cheap flow meters for reading flow from 0 l/min to 100 l/min or from 60 l/min to 300 l/min, but flow meters that cover the entire range 0-300 l/min are extremely more expensive and complex to implement.

[0011] It is, therefore, necessary, if it is desired to obtain readings at both low and high flow rates and at the same time maintain a low price for the final product, to use multiple flow meters or other additional systems and equipment that are able to guarantee measurement in both conditions.

[0012] Even the most recent pumps equipped with electronic control and flow estimation are not able to provide indications or measurements of the flow at both high and low flow rates and, in particular, in measurements at low flow rates, they make numerous measurement errors.

[0013] The increase in the tools needed to extend the range entails some possible disadvantages including: [0014] the increase in costs due to greater instrumentation; [0015] the increase in complexity of the hydraulic systems on which flow measurements must be carried out; [0016] the need to carry out maintenance on multiple instruments with the consequent increase in downtime for the system and maintenance costs; [0017] the presence of unwanted pressure losses due to the fact of having to insert an instrument for low flow rates into a duct for high flow rates; [0018] the most complex and difficult reading and analysis of data.

SUMMARY OF THE INVENTION

[0019] The purpose of the present invention is therefore to provide a flow meter that allows resolving the inconveniences and critical issues mentioned above.

[0020] In particular, the aim of the present invention is to create a flow meter that allows, for example in combination with an electronic or electromechanical pump, the measurement of the flow rate of a fluid at low flow rates and the estimation, by the pump, of the same fluid at high flow rates, limiting pressure losses.

[0021] Another purpose of the present invention is to provide a flow meter for the measurement at high and low flow rates of a fluid that passes through it, extending the measurement ranges compared to the flow meters present in the prior art.

[0022] A further purpose of the present invention is to provide a flow meter for measurement at high and low flow rates which does not require the use of a second flow meter or other instrumentation.

[0023] Another object of the present invention is to provide a flow meter for measurement at high and low flow rates, which does not require increasing the complexity of the hydraulic systems.

[0024] A further object of the present invention is to provide a flow meter for measurement at high and low flow rates which requires minimal maintenance.

[0025] Another object of the present invention is to provide a flow meter that is simple to set up.

[0026] A further purpose of the present invention is to realize a flow meter for measuring high and low flow rates that is economical to manufacture and simple to install by virtue of the advantages achieved.

[0027] These and other purposes are achieved by a flow meter according to the attached independent claim.

[0028] Further detailed technical features are reported in the attached dependent claims.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0029] The present invention will be now described, for illustrative but not limitative purposes, according to its preferred embodiments, with particular reference to the figures of the enclosed drawings, wherein:

[0030] FIGS. 1A-1C are perspective views of different embodiments of a flow meter with parallel bypass according to the invention;

[0031] FIGS. 1D-1E are perspective views of different embodiments of a flow meter with parallel bypass and reinforcement structure according to the invention;

[0032] FIGS. 2A-2B are sectional views of different embodiments of a flow meter with bypass and Venturi tube according to the invention;

[0033] FIGS. 3A-3B are sectional views of different embodiments of a flow meter with coaxial bypass and axial impeller, having respectively parallel axis magnets (FIG. 3A) and radial axis magnets (FIG. 3B);

[0034] FIG. 4 is a perspective view of the flow meter with coaxial bypass of FIG. 3B;

[0035] FIGS. 5A-5B are sectional views illustrating the operation respectively at high (5A) and low (5B) flow rates of the flow meter of FIG. 3B;

[0036] FIGS. 6A-6B are perspective sectional views of the flow meter with coaxial bypass in the embodiment with two axial impellers;

[0037] FIG. 7 is a graph that compares the pressure drops, as the flow rate varies, of the flow meters of FIGS. 1D and 1E;

[0038] FIG. 8 is a graph that compares the pressure drops and the linearity of the rpm, as the flow rate varies, of the flow meters with bypass of the embodiments visible in FIGS. 1D, 1E, and with

only the axial impeller inserted in a 20 mm duct.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

[0039] With reference to the figures mentioned, preferred embodiments of a flow meter are represented with bypass according to the invention.

[0040] Referring to the figures above, the flow meter object of the invention is indicated with the reference numeral **100** and, in a first embodiment, according to the present invention, comprises a casing **1**, which has two different ducts **11**, **12**, in which a fluid flows, whose flow rate has to be measured.

[0041] In particular, the casing **1** has a first duct **11** or bypass duct, in which the fluid flows at low flow rates, and a second duct **12**, in which the fluid flows at high flow rates.

[0042] The casing **1** of the flow meter **100** has a first end **101**, into which the fluid enters, and a second end **102**, from which the same fluid exits.

[0043] In preferred embodiments, the ends **101** and **102** have threads for connecting the flow meter **100** to other elements of a plumbing system, such as pumps or pipes.

[0044] Advantageously, if the fluid has a low flow rate, it passes through the first bypass duct **11** and the flow meter **100** measures the flow rate while, if the fluid has a high flow rate, it passes through the second duct **12**, and the measurement or estimate the flow rate is typically entrusted to an electromechanical pump equipped with electronics for measuring the fluid at high flow rates, which can be positioned downstream or upstream of the flow meter **100**.

[0045] In particular, the electronic or electromechanical pump can be connected directly or indirectly, for example by means of ducts or pipes, to the flow meter **100**.

[0046] Advantageously, the combination of flow meter **100** and electromechanical pump allows good flow measurement at both high and low flow rates.

[0047] In fact, the measurement at low flow rates is entrusted to the flow meter **100** (the electromechanical pump at low flow rates is not capable of carrying out precise measurements) while, at high flow rates, the estimate or measurement is entrusted to the electromechanical pump since the flow meter, in some of its embodiments, may not be able to estimate it correctly.

[0048] In some embodiments, as will be better illustrated below, in the case of the presence of a high flow rate fluid, the crossing occurs both through the second duct **12** and through the first duct **11**.

[0049] In a generic embodiment, the flow meter **100** also comprises: [0050] a measuring element **2** for measuring the flow at low flow rates and arranged inside the first duct **11**; [0051] a closing element **3** adapted to prevent the passage of the flow in one of the two ducts **11**, **12** depending on the flow rate; [0052] an elastic element **4**, connected to the closing element **3** and capable of bringing it back into position after it has been subjected to a high flow rate flow.

[0053] In preferred embodiments, the flow meter **100** also comprises a deflector **7** to convey the flow in the direction of the first duct **11**.

[0054] More in detail, if the flow is at low flow rates, the closing element **3** keeps the duct **12** closed and the fluid can exit the casing **1** only through the duct **11**.

[0055] When the flow is at high flow rates, the pressure of the liquid overcomes the resistance of the elastic element **4**, which kept the closing element **3** in position, and closed the second duct **12**, moving the same closing element **3**, which can operate in closing on the first duct **11**.

[0056] Advantageously, when the fluid has overcome the resistance of the elastic element **4**, the closing element **3** closes, totally or partially, the first duct **11** and the fluid can exit only from the second duct **12** or from both ducts **11** and **12**.

[0057] Subsequently, if the flow returns to having low flow rates, no longer being able to overcome the resistance of the elastic element **4**, the latter moves the closing element **3** bringing it to its initial position, to close the second duct **12** and let the fluid come out only from the first duct **11**.

[0058] In other words, the elastic element **4** assumes a rest or closing configuration, in which the closing element **3** closes the second duct **12** and the pressure of the fluid does not overcome its

resistance and a stressed or opening configuration, in which the closing element **3** opens the second duct **12**, and the pressure of the fluid is able to overcome its resistance.

[0059] With reference to the embodiments visible in FIGS. **1A-1E**, a flow meter **100** with parallel bypass is visible.

[0060] In the embodiments of FIGS. **1A-1E**, the measuring element **2** is an impeller, by way of example a radial impeller, positioned inside the first bypass duct **11**.

[0061] In these embodiments, the first duct **11** and the second duct **12** are placed parallel to each other.

[0062] More in detail, the embodiments illustrated in FIGS. **1A**, **1B**, and **1E** present a bulkhead closing element, while, in the embodiments illustrated in FIGS. **1C** and **1D**, the closing element is a shutter.

[0063] With reference to FIG. **1A**, a first embodiment of the flow meter **100** object of the invention is illustrated, in which the measuring element **2** is a radial impeller, the closing element **3** is a bulkhead and the elastic element is a torsion spring arranged inside the casing **1**.

[0064] In FIG. **1B**, however, a second embodiment of the present invention is visible, in which the elastic element **4**, also in this case a torsion spring, is positioned externally to the casing **1** so that the elastic element **4** itself can be easily adjusted from the outside.

[0065] More in detail, the embodiment of FIG. **1A** is structurally more resistant than the embodiment of FIG. **1B** and has the advantage of obtaining greater water tightness from the connections of the casing **1** and the output of the spring shaft.

[0066] In the embodiments visible in FIGS. **1A** and **1B**, the low flow rate fluid flows in the first duct **11**, causing the measurement element **2** to rotate, which allows the flow rate to be measured. In case of the passage of a fluid at a high flow rate, the flow, overcoming the resistance of the elastic element **4**, moves the closing element **3** (bulkhead), which closes the first duct **11** and allows the fluid to escape from the casing **1** through the second duct **12**.

[0067] With reference to FIG. **1C**, an embodiment of the flow meter **100** is illustrated, in which the closing element **3** is a shutter positioned, together with the elastic element **4**, inside the second duct **12**.

[0068] In this case the elastic element **4** is a compression spring.

[0069] Compared to this third embodiment, the elastic element **4** of the embodiment of FIG. **1A** is more easily calibrated and the flow meter **100** is overall more resistant to the pressure of a high flow rate fluid.

[0070] In FIG. **1D**, a flow meter **100** similar to that of FIG. **1C** is visible, reinforced by means of a reinforcing structure **5**, positioned externally to the casing **1**, designed to increase the pressure resistance of the casing **1**.

[0071] More in detail, the reinforcing structure **5** is positioned externally to the second duct **12**, in which the high-flow fluid flows.

[0072] With reference to FIGS. **1D** and **1E**, the reinforcing structure **5** wraps the ends **101** and **102** of the casing **1** and includes cylindrical elements **51** arranged along the direction of the main extension of the casing **1**.

[0073] Advantageously, the reinforcing structure **5** is also threaded at the ends **101** and **102**. This thread allows the connection to other elements of a hydraulic system such as pumps, pipes, ducts, etc.

[0074] By way of example, the reinforcement structure **5** is made of metallic material.

[0075] In the embodiment of FIG. **1D**, the second duct **12** comes into operation with high-flow fluids, which allow the shutter to open.

[0076] Compared to this last embodiment, that of FIG. **1A**, advantageously, allows a total conveyance of the flow through the second duct **12** when the closing element **3** is open and the flow that passes through the first duct **11** is more silent.

[0077] With reference to FIG. **1E**, a fifth embodiment of the flow meter **100** with parallel bypass is

visible, in which, unlike the embodiment of FIG. 1D, the closing element **3** is not a shutter but is a bulkhead and the elastic element **4** is not a compression spring but a torsion spring.

[0078] In this embodiment, in the case of high flow rate, the reinforcing structure demonstrates its effectiveness. In fact, the tilting bulkhead comes into operation correctly by opening the bypass (second duct **12**).

[0079] The embodiments with bulkhead closing element **3** remain preferred to the other embodiments described so far, as they guarantee the complete opening of the second duct **12**, which involves less noise.

[0080] Furthermore, the flow meter **100** of FIG. 1E, as well as the other bulkhead flow meters **100**, has excellent performance both in relation to pressure drops and in terms of noise.

[0081] With reference to FIGS. 2A-2B, flow meters **100** are visible, which operate according to the Venturi differential pressure principle.

[0082] In such flow meters **100**, the closing element **3** is a shutter and the elastic element **4** is a compression spring.

[0083] In these embodiments, the measuring element **2** is a differential pressure sensor that reads the low flow rates that pass through at least one Venturi tube **6**. The Venturi tubes **6** are placed in correspondence with the first duct **11**.

[0084] More in detail, the differential pressure sensor (not visible in the figures) is able to read the Delta-P due to the variation in flow speed imposed by the Venturi tubes **6** on the fluid flowing through them.

[0085] Referring to FIG. 2A, the flow meter **100** has a parallel bypass. In fact, the two ducts **11** and **12** are arranged parallel to each other.

[0086] In particular, when the fluid is at a low flow rate, it flows through the Venturi tube **6** of the first duct **11** and the differential pressure sensor measures the flow rate. In case of low flow rate, the closing element **3** (shutter) keeps the second duct **12** closed.

[0087] As the flow rate increases, the growing pressure exerted by the flow manages to overcome the resistance of the elastic element **4** and open the shutter, allowing the flow to also enter the second duct **12**.

[0088] It is necessary to size the Venturi tube **6** in order to impose a minimum Delta-P, due to the minimum sensitivity of the differential pressure sensors, compared to a minimum flow rate.

[0089] At the same time, it is necessary not to reach too high-pressure drops for higher flow rates, for example up to 30 l/min.

[0090] With reference to the embodiment of FIG. 2B, an attempt was made to improve the embodiment of FIG. 2A by increasing the sensitivity of the Venturi tube and arranging two Venturi tubes **6** in parallel.

[0091] The flow meter **100** of FIG. 2B is a coaxial bypass flow meter.

[0092] More in detail, as visible in FIG. 2B, the first duct **11** has two Venturi tubes **6** in parallel, a first convergent tube and a second divergent tube.

[0093] Advantageously, in this situation, it is possible to double the Venturi effect in the first duct **11** and reduce the pressure drops of the closing element **3** compared to the embodiment of FIG. 2A.

[0094] Even more advantageously, in this embodiment, the closing element **3** is a toroidal shutter while the bypass is coaxial since the first duct **11** is arranged coaxially with the second duct **12**.

[0095] Operationally, when the flow is at a low flow rate, it flows in the first duct **11** through the Venturi tubes **6** while, when the flow is at a high flow rate, it pushes the closing element **3** (shutter), which compresses the elastic element **4** allowing the fluid to also pass into the second duct **12**.

[0096] With reference to FIGS. 3A-3B, further embodiments of flow meters **100** with coaxial bypass are visible.

[0097] In such flow meters, the closing element **3** is a toroidal shutter, the elastic element **4** is a compression spring and the measuring element **2** is an axial impeller arranged inside the first duct **11**.

[0098] In these embodiments, the two ducts are arranged coaxially, with the first duct **11** arranged internally and coaxially with the second duct **12**.

[0099] With reference to FIGS. **3A** and **3B**, an embodiment of the flow meter **100** is illustrated, in which the low flow rate flow is conveyed inside the first duct **11** and rotates the impeller, which allows the flow rate to be measured.

[0100] To measure the flow rate in the first duct **11**, two magnets **21** arranged internally thereof are used, which allow the reading of the revolutions per minute (RPM) of the axial impeller and the consequent measurement of the flow rate.

[0101] In particular, the two magnets **21** pass facing a Hall sensor, arranged in a seat present in the first duct **11**, which allows the measurement of the revolutions per minute and consequently allows the determination of the flow rate of the fluid passing through the first duct **11**.

[0102] More in detail, the two magnets **21** can be arranged, always inside the impeller **2**, with a parallel axis with respect to the axial impeller **2** itself (FIG. **3A**) or with a radial axis with respect to the axial impeller **2** itself (FIG. **3B**).

[0103] As the flow rate and therefore the pressure drops increase, the increasing pressure manages to open the closing element **3** (toroidal shutter), which opens the second duct **12** and allows the flow to pass through.

[0104] The embodiments with axial bypass and shutter closure, for example, visible in FIGS. **3A** and **3B**, present excellent behavior in terms of pressure drops, good silence, and good reading of the minimum flow, as well as reduced overall dimensions compared to the previous embodiments.

[0105] Furthermore, the cost is reduced compared to other embodiments, in which a differential pressure sensor is present, and the consequent difficulty in finding such a sensor having the required requirements.

[0106] It is necessary that the rotation speed of the impeller is uniform and that a slowdown is avoided when an impeller blade passes close to the Hall sensor.

[0107] Referring to FIG. **3B**, a flow meter **100** with coaxial bypass and shutter closure is visible.

[0108] This embodiment works operationally like the embodiment of FIG. **3A**, but presents an improvement due to the fact that the obstruction on the first duct **11** due to the seat of the Hall sensor is eliminated.

[0109] This positioning of the Hall sensor caused the slowdown of the impeller mentioned previously, due to the fact that the structure containing it is very bulky and partially obstructs the first duct **11**.

[0110] This improvement is due to the radial positioning of the magnets **21** with respect to the axial impeller **2**.

[0111] In particular, the magnets **21** of the axial impeller **2**, being positioned with a radial axis with respect to the impeller, allow the Hall sensor to be positioned outside the first duct **11**.

[0112] Advantageously, this embodiment allows to obtain lower pressure drops, better silence and uniform rotation speed of the impeller **2**.

[0113] In a further embodiment, the impeller is made of plastoferrite. In this embodiment, due to the characteristics of the material, in which the impeller is made, the presence of the magnets **21** is not necessary.

[0114] In fact, the Hall sensor is able to detect the rotation of the plastoferrite impeller, determining its revolutions per minute and consequently the flow rate of the fluid.

[0115] In FIG. **4** the flow meter with coaxial bypass of FIG. **3B** is visible from the outside.

[0116] Referring to FIGS. **5A** and **5B**, the operation of the flow meter with coaxial bypass of FIG. **3B** at low flow rate (FIG. **5A**) and at high flow rate (FIG. **5B**) is visible.

[0117] In FIG. **5A** it is visible how the flow entering from the first end **101** passes through the flow meter **100** in the first duct **11** and then exits from the second end **102**. During transit in the first duct **11**, the flow causes the measuring element **2** to rotate, which allows the measurement of the flow rate.

[0118] In FIG. 5B it is visible how the flow entering from the first end **101** manages, thanks to its pressure, to push the closing element **3** overcoming the resistance of the elastic element **4**. In this situation, the flow can pass through the flow meter **100** both in the first duct **11** and in the second duct **12** and the estimate of the flow rate is entrusted to an electronic or electromechanical pump. [0119] This electronic or electromechanical pump, equipped with electronics for measuring the fluid at high flow rates, is connected directly or indirectly to the flow meter **100** and positioned downstream or upstream of the same flow meter **100**.

[0120] Advantageously, the flow meter **100** can in fact be installed in a system also comprising an electronic or electromechanical pump, which, via an electronic card present inside it, is capable of making flow estimates at high flow rates.

[0121] Furthermore, with reference to the flow meter **100** with coaxial bypass, it is possible to measure the fluid at high flow rates through the analysis of the increasing monotonic response of the rotation speed of the axial impeller **2**.

[0122] Advantageously, the analysis of this monotonic response occurs by means of the Hall sensor.

[0123] With reference to FIGS. 6A and 6B, an embodiment of the flow meter **100** with coaxial bypass is illustrated, in which there is a second measuring element **22**, in particular a second impeller, for measuring the fluid even at high flow rates.

[0124] As visible in FIGS. 6A and 6B, the second impeller **22** is of the axial type and is advantageously placed upstream with respect to the shutter **3** and the axial impeller **2**.

[0125] The second impeller **22** is configured to move when the second duct **12** is opened by the closing device **3**.

[0126] Advantageously, in this embodiment, the total flow measurement is obtained by combining the data obtained from the axial impeller **2** and the data obtained from the impeller **22**.

[0127] Advantageously, the rotation of each impeller is detected by a respective Hall sensor.

[0128] Preferably, the data from the two impellers are analyzed, and combined, and linked by an electronic control.

[0129] Still preferably, the combination of the measurement of a single impeller **2** with bypass and estimation of the electronic pump is also entrusted to an electronic control, which, in addition to estimating the flow rates, carries out the correct combination of the two flow rate ranges.

[0130] Advantageously, the embodiment with two impellers allows the flow meter **100** to measure a fluid at both high and low flow rates.

[0131] Even more advantageously, it is possible to have a single flow meter **100** capable of measuring the flow at low flow rates, at high flow rates, and in intermediate ranges, extending the measurement range compared to known instruments.

[0132] To analyze and evaluate the performance of the different embodiments of the flow meters **100**, tests were carried out, in which some of the embodiments previously described were compared.

[0133] It should be noted that these tests were carried out in the laboratory with particular instruments and components of the flow meters and can be used as a comparison of the different embodiments under the same conditions of use.

[0134] A first test, the results of which can be seen in FIG. 7, compared the pressure drops (in meters) in relation to the flow rate (in l/min) of some different flow meters. The embodiments compared were those of FIG. 1D, 1E.

[0135] From a reading of the graph, it can be seen that the embodiment of FIG. 1E has lower pressure losses compared to the embodiment of FIG. 1D.

[0136] In further tests, the threshold at which the flow rate (in l/min) allows the opening of the closing element **3** and the passage of the fluid into the second duct **12** was analyzed.

[0137] These tests were carried out for the embodiments of FIGS. 1D and 1E.

[0138] Regarding the embodiment of FIG. 1D, the following data were obtained during a test run.

TABLE-US-00001 Flow Meter of FIG. 1D Q p1 p2 Delta p [l/min] [bar] [bar] [bar] H [m] 0 -0.058
-0.080 0.022 n/d 5 -0.048 -0.069 0.021 -0.01 15 -0.003 -0.034 0.031 0.09 20 0.027 -0.022
0.049 0.28 40 0.060 -0.008 0.068 0.47 60 0.070 0.000 0.070 0.49 85 0.031 -0.044 0.075 0.54 95
0.026 -0.061 0.087 0.66 100 0.087 -0.008 0.095 0.74 120 0.087 -0.053 0.140 1.20 140 0.130
-0.050 0.180 1.61 170 0.200 -0.047 0.247 2.30 200 0.280 -0.040 0.320 3.04 230 0.404 -0.030
0.434 4.20 260 0.524 -0.014 0.538 5.26 297 0.686 0.000 0.686 6.77

[0139] In particular, it was possible to notice how at the flow rate of 85 l/min, the limit for the movement of the closing element **3** for the opening of the second duct **12** occurred.

[0140] It was then analyzed how at the flow rate of 95 l/min the closing element **3** is in a partially open position and how at the flow rate of 100 l/min the closing element **3** is in a completely open position.

[0141] Regarding the embodiment of FIG. **1E**, in one test run, the following data were obtained.

TABLE-US-00002 Flow Meter of FIG. 1E Q p1 p2 Delta p [l/min] [bar] [bar] [bar] H [m] 0 0.001
-0.018 0.019 n/d 5 0.023 0.001 0.022 0.03 15 0.046 0.013 0.033 0.14 20 0.062 0.019 0.043 0.24 30
0.036 -0.019 0.055 0.37 40 0.095 0.035 0.060 0.42 60 0.107 0.047 0.060 0.42 80 0.110 0.047
0.063 0.45 100 0.113 0.036 0.077 0.59 120 0.086 0.003 0.083 0.65 140 0.029 -0.055 0.084 0.66
170 0.068 -0.020 0.088 0.70 200 0.070 -0.040 0.110 0.93 230 0.098 -0.030 0.128 1.11 260 0.128
-0.019 0.147 1.31 304 0.205 0.002 0.203 1.88

[0142] In particular, it was possible to notice how at the flow rate of 30 l/min the limit for the movement of the closing element **3** for the opening of the second duct **12** occurred.

[0143] It was then analyzed that at the flow rate of 95 l/min the closing element **3**, in this case a bulkhead, is in a partially open position (approximately at) 45° and that in this embodiment the closing element **3** is never in a completely open position, not even at maximum capacity.

[0144] Regarding the embodiment of FIG. **3A**, the following data have been obtained.

TABLE-US-00003 Flow Meter of FIG. 3A Q p2 Delta p [l/min] p1 [bar] [bar] [bar] H [m] RPM 0
-0.046 -0.075 0.029 n/d 1 n/d n/d n/d n/d 16 3 n/d n/d n/d n/d 138 5 -0.045 -0.075 0.030 0.01 240
10 -0.043 -0.074 0.031 0.02 450 15 -0.041 -0.074 0.033 0.04 700 20 -0.035 -0.074 0.039 0.10
845 30 -0.020 -0.074 0.054 0.26 1200 40 0.000 -0.074 0.074 0.46 1482 50 0.014 -0.074 0.088
0.60 1578 55 0.015 -0.074 0.089 0.61 1464 60 0.005 -0.074 0.079 0.51 1420 85 0.030 -0.070
0.100 0.72 1560 95 0.037 -0.068 0.105 0.78 1656 100 0.031 -0.066 0.097 0.69 1668 120 0.038
-0.063 0.101 0.73 1740 140 0.045 -0.057 0.102 0.74 1818 170 0.058 -0.050 0.108 0.81 1932 200
0.074 -0.038 0.112 0.85 2040 230 0.090 -0.028 0.118 0.91 2196 260 0.112 -0.012 0.124 0.97
2340 305 0.187 0.008 0.179 1.53 2640

[0145] In particular, it was possible to verify that the limit for the movement of the closing element **3** for the opening of the second duct **12** is at the flow rate of 60 l/min.

[0146] The results visible in the previous tables were detected using a flow meter **100** having an elastic element **4** of the compression spring type CO 1.8×41×80×8 of 10N, two magnets **21** of diameter 2 mm and height 3 mm of the attractive magnetic force of approx. 160 g each, arranged one per blade and with reversed polarities.

[0147] Revolutions per minute (RPM) were detected with Hall sensor TLV49681TAKBKA1-ND and readings were taken with an oscilloscope.

[0148] The following tests have made it possible to identify how the embodiments with coaxial bypass and axial impeller lead to obtaining better results and performances compared to other embodiments.

[0149] Referring to FIG. **8**, a graph is shown that compares the pressure drop and linearity curves in relation to the flow rate. Linearity is evaluated as the ratio of frequency in revolutions per minute (RPM) to flow rate in l/min (Q).

[0150] This graph relates to the embodiment of the flow meter **100** with coaxial bypass, in which the fluid at high flow rates is measured through the analysis of the increasing monotonic response of the rotation speed of the axial impeller **2**.

[0151] The graph of FIG. 8 compares the curves relating to the embodiments with radial axis magnets (FIG. 3B), with parallel axis magnets (FIG. 3A) and to an embodiment, in which the casing **1** of the flow meter is simply a cylindrical duct with a diameter of 20 mm, without bypass, inside which there is only an axial impeller and no other components.

[0152] In particular, with reference to the embodiment of FIG. 3A, it is visible how the linearity at low flow rates (up to 50 l/min) is almost close to an inclined straight line representable with $y=60.197x+29.39$ and $R^2=0.9967$.

[0153] From the graph in FIG. 8 it is also clearly visible that: [0154] in the embodiment in which only the axial impeller is present in the cylindrical duct, the linearity and pressure drops increase very quickly even at low flow rates and are considerably greater than in other embodiments for flow rates greater than 50l/min; [0155] in the embodiments with bypass there is a more accentuated inclination of the curve up to a flow rate of 50l/min and less accentuated for higher flow rates; [0156] the embodiment with radial magnets presents much greater linearity than the embodiment with parallel axis magnets, for all flow rates.

[0157] Advantageously, it is clear how the flow meters **100** with coaxial bypass (embodiments of FIGS. 3A and 3B) are capable of measuring the flow rate in several decades and even up to 300 l/min.

[0158] Again advantageously, since the flow meter **100** is capable of obtaining the revolutions per minute of the measuring elements **2** and **22**, it is possible to obtain the flow rate since the latter can be characterized a priori on the basis of the revolutions per minute of the measuring elements **2** and **22**.

[0159] More in detail, by knowing a priori which flow rate corresponds to a certain value of revolutions per minute of the measuring elements **2** and **22**, it is possible to establish the value once the flow passes through the flow meter **100**.

[0160] From the description given, the characteristics of the flow meter with bypass, object of the invention, just as the advantages, are clear.

[0161] Finally, it is clear that numerous other variations can be made to the system in question, without departing from the principles of novelty inherent in the inventive idea, just as it is clear that, in the practical implementation of the invention, the materials, shapes and the dimensions of the illustrated details may be any according to needs and they may be replaced with other equivalent ones.

[0162] Where the features and techniques mentioned in any claims are followed by reference marks, such reference marks have been included for the sole purpose of increasing the intelligibility of the claims and, accordingly, such reference marks have no limiting effect on the interpretation of each element identified by way of example by these reference signs.

Claims

1. A flow meter for measuring the flow rate of a fluid characterized by comprising: a casing having a first end into which the fluid enters and a second end from which the fluid exits, and within which a first duct for receiving the fluid is present when it is at low flow rates; a measuring element arranged internally to the first duct for measuring the flow within the first duct; said flow meter being characterized by also comprising: a second duct, placed inside the casing, for receiving the fluid at high flow rates; a closing element which, when the flow rate exceeds an opening threshold, allows the fluid to flow into the second duct; an elastic element, connected to the closing element, for moving the closing element from a closing configuration, in which the flow rate does not exceed the opening threshold, not overcoming the resistance of the elastic element and the closing element closes the second duct, to an opening configuration in which the flow rate exceeds the opening threshold, overcoming the resistance of the elastic element, and the closing element opens the second duc.

2. Flow meter according to claim 1, wherein when the flow rate exceeds the opening threshold, the closing element (3) closes at least partially the first duct and allows the fluid to flow, totally or partially, into the second duct.
 3. Flow meter according to claim 1, wherein the first end and the second end are threaded.
 4. Flow meter according to claim 1, wherein the first duct is arranged internally and coaxially with respect to said second duct.
 5. Flow meter according to claim 1, wherein said measuring element is an axial impeller.
 6. Flow meter according to claim 5, wherein said measuring element is made of plastroferrite.
 7. Flow meter according to claim 1, wherein said closing element is a toroidal shutter.
 8. Flow meter according to claim 1, wherein said elastic element is a compression spring.
 9. Flow meter according to claim 1 characterized by comprising: one or more magnets arranged internally to the measuring element, at least one sensor configured to detect the transit of the magnets and determine the flow rate within the first duct.
 10. Flow meter according to claim 9, wherein said magnets are arranged in a radial or axial position with respect to said measuring element.
 11. Flow meter according to claim 9, wherein said sensor is a Hall sensor.
 12. Flow meter according to claim 1, characterized by comprising an additional measuring element, placed upstream of the measuring element, suitable for measuring the flow of the fluid at high flow rates in case of opening of the second duct by the closing device.
 13. A plant for measuring the flow of a fluid comprising: a flow meter according to claim 1 for measuring fluid at low flow rates; an electronic or electromechanical pump equipped with electronics for measuring the flow of the fluid at high flow rates, connected directly or indirectly to the flow meter, for measuring the flow of the fluid at high flow rates.
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